

Project 1

I am assuming this structure:

```
Part2-SHP-RTSSP/
|
|— secureStreamServer/
|   |— secureStreamServer.java
|   |— RTSSP.java
|   |— CryptoConfig.java
|   |— SHP.java
|   |— SHPContext.java
|   |— CertificateUtils.java
|   |— movies/
|       |— cars.dat
|
|— secureUDProxy/
|   |— secureUDProxy.java
|   |— RTSSP.java
|   |— CryptoConfig.java
|   |— SHP.java
|   |— SHPContext.java
|   |— CertificateUtils.java
|   |— config.properties
|
|— Cryptoconfig.conf
```

1. Open terminal in the project root

Go to:

```
cd C:\Users\alexa\Documents\GitHub\criptographyProject\SpecialRealTimeMediaStreaming\Part2-SHP-RTSSP
```

You should be inside:

```
Part2-SHP-RTSSP
```

Not inside `secureStreamServer` or `secureUDProxy` .

2. Check `config.properties`

Open:

```
secureUDProxy/config.properties
```

It should look like this:

```
remote=localhost:8888
localdelivery=localhost:7777

servercontrol=localhost:9999
movie=cars.dat
```

```
supportedCiphersuites=AES/GCM/NoPadding,ChaCha20-Poly1305,AES/CTR/NoPadding
```

Meaning:

```
servercontrol=localhost:9999 → SHP handshake port
remote=localhost:8888      → proxy receives encrypted RTSP stream here
localdelivery=localhost:7777 → proxy forwards decrypted video to VLC here
movie=cars.dat             → requested movie
```

3. Make sure the movie exists

You need:

```
secureStreamServer/movies/cars.dat
```

If your movie is somewhere else, either move it there or change this line in the server code:

```
StringmovieFile="secureStreamServer/movies/"+movieName;
```

4. Generate the keystores and truststores

Run these commands from the **project root**.

Server certificate

```
keytool-genkeypair-alias server-keyalg EC-groupname secp256r1-sigalg SHA256withECDSA-keystore secureStreamServer/server-keystore.j
ks-storepass changeit-keypass changeit-dname"CN=StreamingServer"-validity365
```

```
keytool-exportcert-alias server-keystore secureStreamServer/server-keystore.jks-storepass changeit-file server.cer
```

Proxy certificate

```
keytool-genkeypair-alias proxy-keyalg EC-groupname secp256r1-sigalg SHA256withECDSA-keystore secureUDPProxy/proxy-keystore.jks-sto
repass changeit-keypass changeit-dname"CN=ProxyBox"-validity365
```

```
keytool-exportcert-alias proxy-keystore secureUDPProxy/proxy-keystore.jks-storepass changeit-file proxy.cer
```

Server trusts proxy

```
keytool-importcert-alias proxy-file proxy.cer-keystore secureStreamServer/server-truststore.jks-storepass changeit-noprompt
```

Proxy trusts server

```
keytool-importcert-alias server-file server.cer-keystore secureUDPproxy/proxy-truststore.jks-storepass changeit-noprompt
```

After this, you should have:

```
secureStreamServer/server-keystore.jks
secureStreamServer/server-truststore.jks

secureUDPproxy/proxy-keystore.jks
secureUDPproxy/proxy-truststore.jks
```

You only need to generate these once.

5. Delete old `.class` files

From the project root:

```
cd secureStreamServer
del *.class
cd ..
```

```
cd secureUDPproxy
del *.class
cd ..
```

6. Compile the server

From the project root:

```
cd secureStreamServer
javac *.java
cd ..
```

7. Compile the proxy

From the project root:

```
cd secureUDPproxy
javac *.java
cd ..
```

8. Open VLC

In VLC:

```
Media → Open Network Stream
```

Use:

```
udp://@:7777
```

Then click **Play**.

Keep VLC open before starting the proxy.

9. Start the server

Open Terminal 1 in the project root:

```
java -cp secureStreamServer secureStreamServer9999
```

Expected output:

```
Server waiting for signed SHP CLIENT_HELLO on port 9999
```

10. Start the proxy

Open Terminal 2 in the project root:

```
java secureUDPproxy.secureUDPproxy
```

Expected proxy output:

```
Proxy listening for RTSSP stream on localhost:8888
Starting SHP handshake with mutual authentication...
SHP CLIENT_HELLO sent and signed
SHP SERVER_HELLO received
Server certificate trusted
SERVER_HELLO signature verified
Selected CipherSuite: AES/GCM/NoPadding
Session keys derived
SHP CLIENT_FINAL sent encrypted
SHP handshake completed
Waiting for RTSSP stream...
```

Expected server output:

```
SHP CLIENT_HELLO received
Proxy certificate trusted
CLIENT_HELLO signature verified
SHP SERVER_HELLO sent and signed
Selected CipherSuite: AES/GCM/NoPadding
Session keys derived
SHP CLIENT_FINAL received and verified
SHP handshake completed with mutual authentication
Starting RTSSP stream...
RTSSP START sent
::::::::::::::::::::
RTSSP END sent
```

Expected proxy output during stream:

```
RTSPS START received: START:cars.dat
.....
RTSPS END received: END:...
Frames received: ...
Packets dropped: 0
```

11. Correct order every time

Use this order:

1. Open VLC on `udp://@:7777`
2. Start server
3. Start proxy